MODULE 06 · DATA

Data Science Basics

A Custom AI-Generated Companion Guide

Turn raw numbers into real insight — the skillset behind every 'data-driven' decision.

DIGITAL GIRLS INITIATIVE PAKISTAN
Empowering young women through free tech education

8–10 weeks · 5–7 hrs/week

What's Inside

01 Welcome & How to Use This Guide
02 Phase 01 — Foundations
03 Phase 02 — Python for Data
04 Phase 03 — Visualization & Analysis
05 Phase 04 — Intro to Machine Learning
06 Free Video Courses
07 Your Practice Checklist
08 A Note From DG·IP

Welcome & How to Use This Guide

Data science is the practice of asking questions of data and finding honest answers — a skill used everywhere from public health to business to sports. This track starts with statistics fundamentals, moves through Python's most-used data libraries, and ends with a first taste of machine learning.

> **Mentor Tip**
> Always plot your data before you model it — a five-second chart often reveals what an hour of code would miss.

Estimated time: 8–10 weeks · 5–7 hrs/week · Level: Basic Python recommended

01 Foundations

Thinking like a data scientist

Before any code, this phase builds the statistical intuition that makes every later tool make sense.

01 What data science actually is
The difference between data analysis, data science, and machine learning.

02 Statistics basics
Mean, median, mode, and standard deviation — the vocabulary of every dataset.

03 From spreadsheets to code
Why code eventually beats spreadsheets for serious analysis.

04 Asking good questions
The most underrated data science skill: framing a clear, answerable question.

★ Practice Project

Take any spreadsheet of data you have access to and write down five questions you could ask of it.

02 Python for Data

The core toolkit

Two libraries — NumPy and pandas — do the vast majority of real-world data work in Python.

01 NumPy basics
Fast numerical arrays, the foundation everything else is built on.

02 pandas DataFrames
Loading, viewing, and slicing tabular data like a spreadsheet in code.

03 Cleaning messy data
Handling missing values, duplicates, and inconsistent formatting.

04 Grouping & aggregating
Summarizing data by category — the pandas equivalent of a pivot table.

★ Practice Project
Load a public CSV dataset (e.g. from Kaggle) and produce a clean summary table by category.

03 Visualization & Analysis

Making data speak

A great chart communicates an insight in seconds — this phase teaches you to build ones that do.

01 Matplotlib basics
Line charts, bar charts, and histograms from scratch.

02 Seaborn for polish
Statistically-aware charts with far less code.

03 Exploratory data analysis
A repeatable process for getting to know any new dataset.

04 Drawing honest conclusions
Avoiding the most common ways charts mislead, even accidentally.

★ Practice Project

Produce a 3-chart mini-report on a dataset of your choice, with one written insight per chart.

04 Intro to Machine Learning

Your first predictive model

This phase is a gentle first step into machine learning — enough to understand what it is and build one real model.

01 Core ML concepts
Training data, features, and predictions, explained without the jargon.

02 scikit-learn basics
The standard, beginner-friendly Python ML library.

03 Your first model
Training a simple regression or classification model.

04 Evaluating a model
Understanding whether your model is actually any good.

★ Practice Project

Train a simple model to predict a value in a beginner Kaggle dataset (e.g. house prices or Titanic survival).

Free Video Courses

Curated, highly-rated, and completely free. Pick whichever teaching style fits you best.

> **freeCodeCamp — Data Analysis with Python**
> Full certification, pandas and NumPy from scratch
> *https://www.freecodecamp.org/learn/data-analysis-with-python/*

> **Kaggle Learn — Micro-Courses**
> Bite-sized, hands-on lessons directly in your browser
> *https://www.kaggle.com/learn*

> **Khan Academy — Statistics & Probability**
> The clearest free foundation in the underlying math
> *https://www.khanacademy.org/math/statistics-probability*

> **StatQuest with Josh Starmer**
> Statistics and ML concepts explained with unmatched clarity
> *https://www.youtube.com/@statquest*

Your Practice Checklist

Print this page or copy it into a notebook. Check off each project as you finish it — real, working projects are what turn this guide into a portfolio.

- [ ] **Phase 01: Foundations — Take any spreadsheet of data you have access to and write down five qu**
- [ ] **Phase 02: Python for Data — Load a public CSV dataset (e.g. from Kaggle) and produce a clean summa**
- [ ] **Phase 03: Visualization & Analysis — Produce a 3-chart mini-report on a dataset of your choice, with on**
- [ ] **Phase 04: Intro to Machine Learning — Train a simple model to predict a value in a beginner Kaggle dat**

Finished all four phases?

Bring your projects to a DG·IP mentor session — we review portfolios and help you plan your next module.

You've got the roadmap.
Now go build something.

This guide is one part of the free DG·IP curriculum. Come back to the website for the live version of this module, curated videos, and seven other free skill tracks.

digitalgirlsinitiative.pk

DIGITAL GIRLS INITIATIVE PAKISTAN

Empowering young women through free tech education